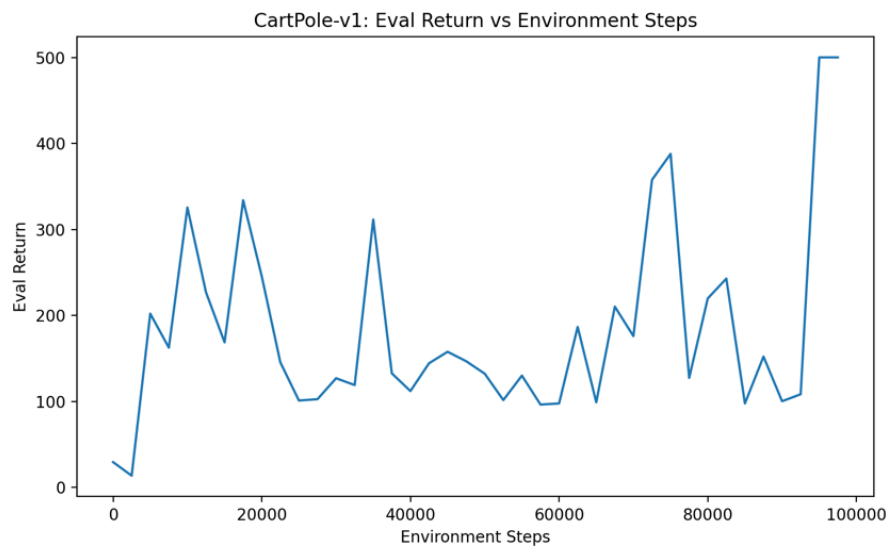


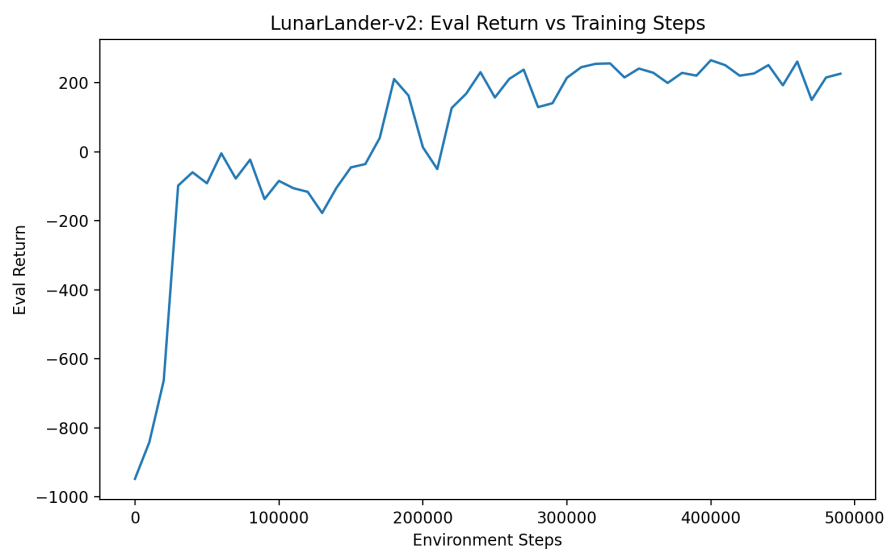
Basic Q-Learning

1. Training Plot

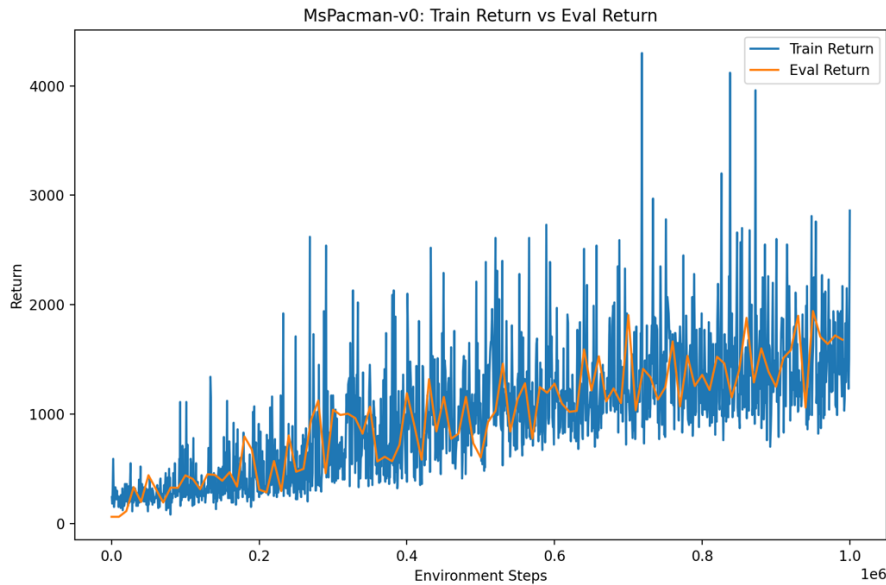


Double Q-Learning

1. LunarLander-v2 training plot



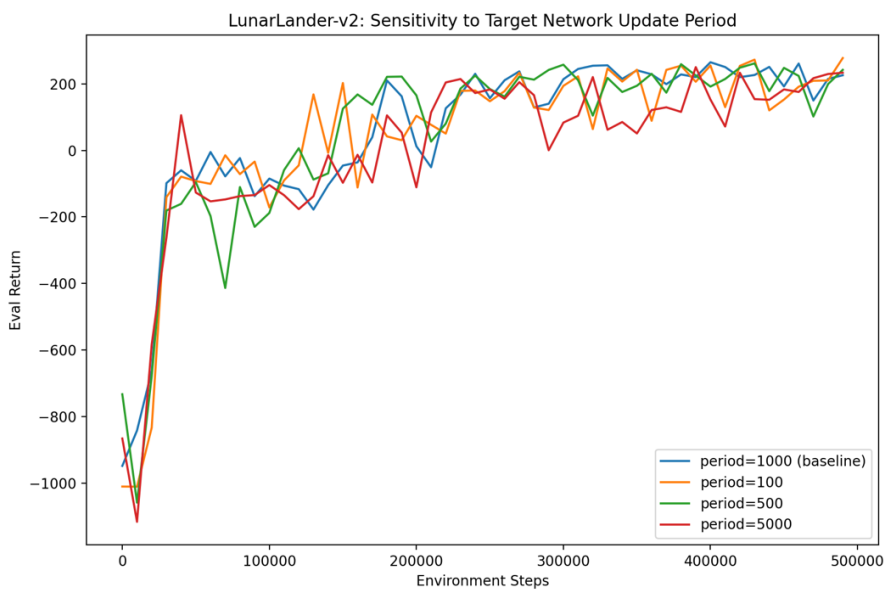
2. MsPacman-v0 training plot



The training return is much noisier than the evaluation returns because it is measured from individual training episodes under an exploratory policy, so it is strongly affected by epsilon-greedy randomness and episode-level variance. In contrast, the evaluation return is computed using the current policy with much less exploration and is usually averaged over multiple trajectories, so it is smoother and better reflects the actual policy quality. This is why the two curves can look quite different, especially early in training.

Experimenting With Hyperparameters

1. Result plot



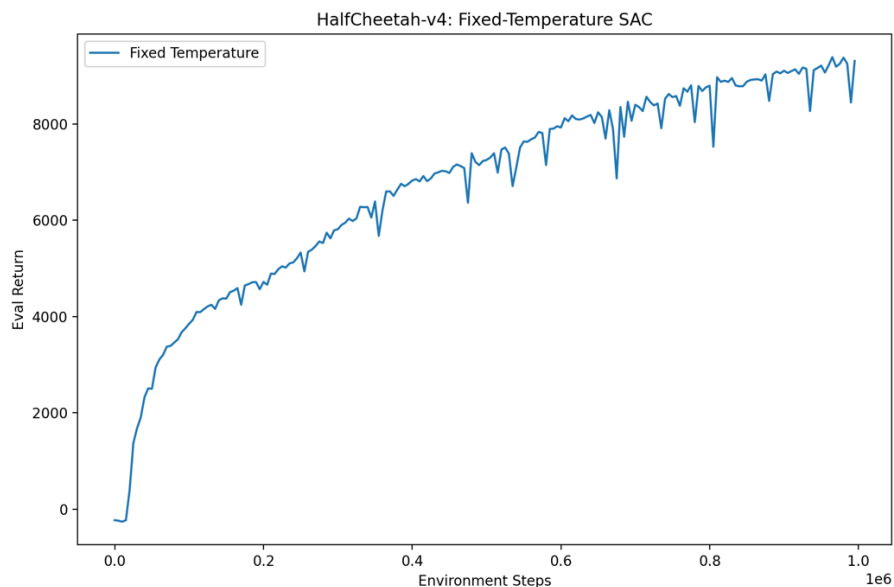
2. Explanation:

I chose to tune the target network update frequency because it directly affects the

stability of Bellman targets during training. From the graph, all four settings eventually achieve convergence to a reasonably good return, which suggests that the basic DQN implementation is working properly. However, when the update period is too small, such as 100, the target network is refreshed too frequently, so the target changes more rapidly and the stabilizing effect of the target network is weakened. This likely explains the larger fluctuations in the orange curve. In contrast, when the update period is too large, such as 5000, the target network lags too far behind the online network, making the targets overly stale. As a result, learning is slower in the early stage, and the performance is somewhat more oscillatory. Overall, the intermediate settings of 500 and 1000 provide a better balance and appear to give more stable learning behavior.

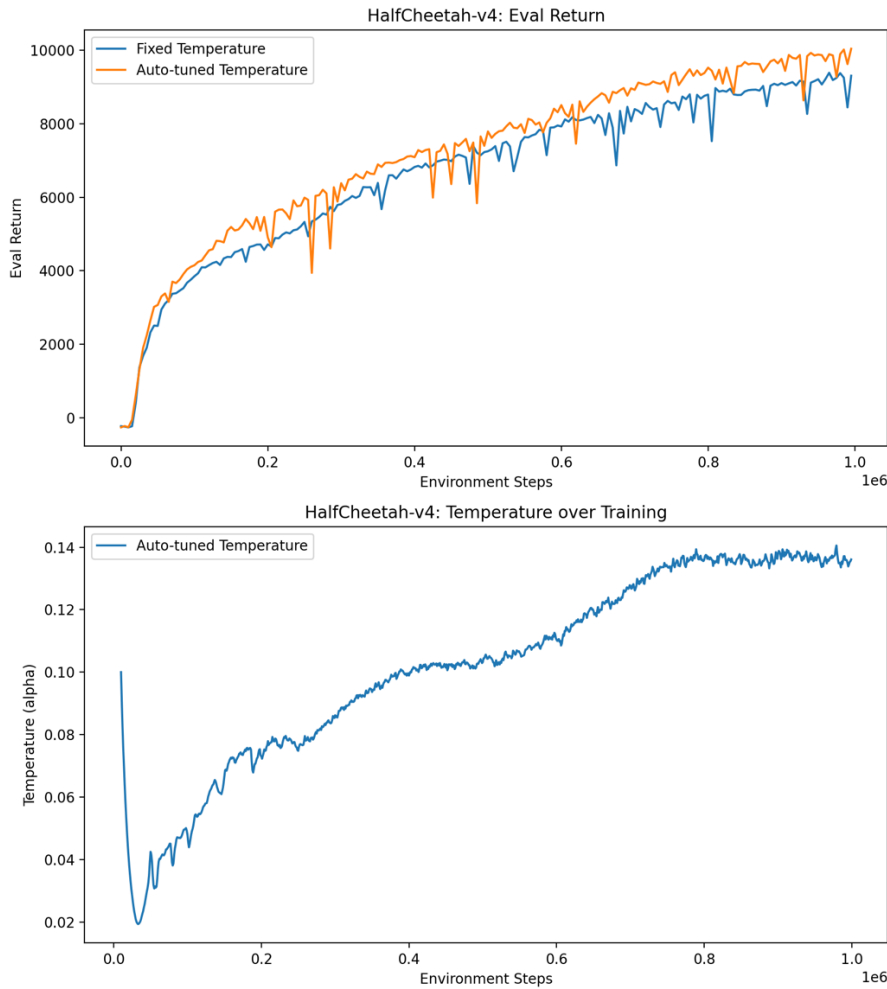
Fixed temperature SAC

1. Training plot



SAC with automatic temperature tuning

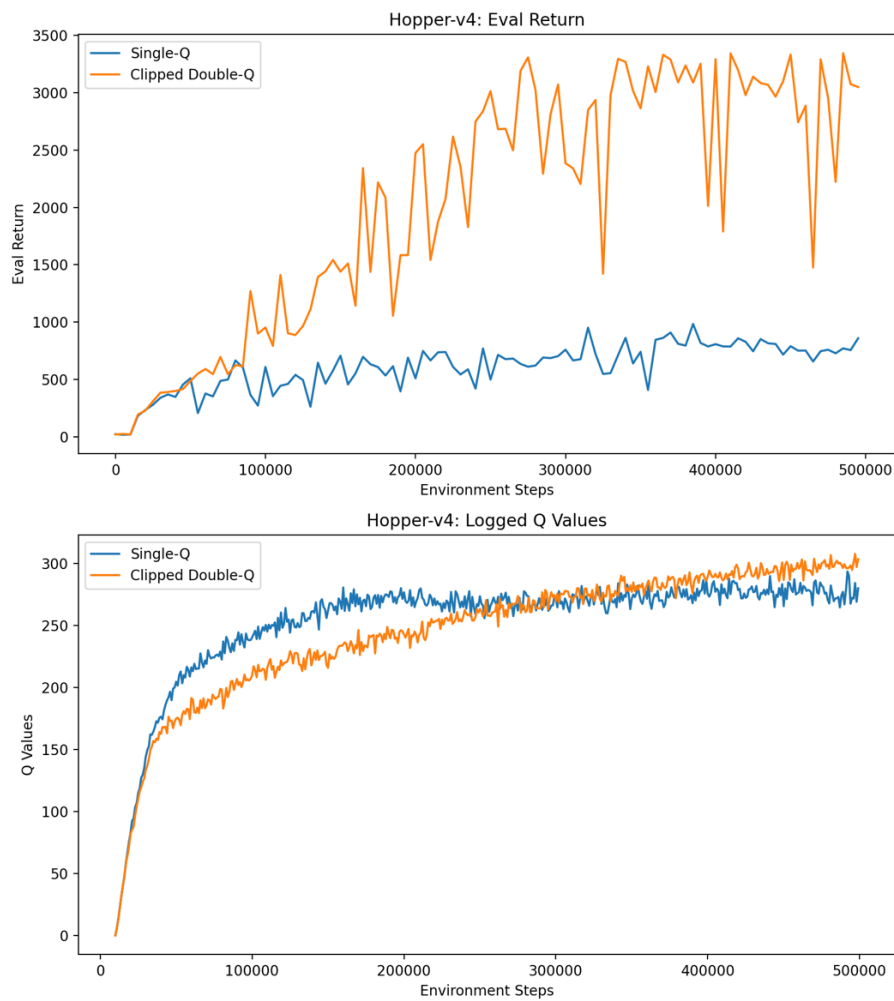
1. Training plot



Auto-tuning achieves at least comparable performance, and in this run it slightly outperforms the fixed-temperature baseline, especially in the later stage of training. The temperature first decreases sharply at the beginning of training, then gradually increases, and finally stabilizes around 0.13–0.14. This suggests that the desired exploration level changes over training. Early on, the algorithm quickly reduces the entropy weight, possibly because the initial policy is already sufficiently random. Later, as learning progresses, the policy becomes more deterministic, and the temperature increases to maintain the target entropy and preserve useful exploration in the high-dimensional continuous control setting. Eventually it stabilizes once the policy entropy and target entropy level become well matched.

Stabilizing Target Values

1. Training plot



2. Analysis

In this experiment, clipped double-Q clearly performs better than single-Q on Hopper-v4. The clipped double-Q agent reaches returns well above 1500 and eventually stays around 2500–3300 for much of training, while the single-Q baseline remains below 1000. This is consistent with the motivation of clipped double-Q: using the minimum of two target critics reduces overestimation bias and provides a more stable learning signal. Although the logged Q-values of the two methods are not dramatically different in magnitude, the single-Q agent achieves much worse policy performance, which suggests that its value estimates are less reliable (larger over estimation bias). In contrast, clipped double-Q produces better returns, indicating that its target values are more stable and better aligned with actual policy quality.